CSST - Complication Signal Surveillance Template

Review identification, purpose and sources

v2.0

Study / protocol	CIP version	Site / code	CSST review ID / revision
Review type ☐ Scheduled ☐ Triggered	**Review period / data cut-off**	**Device / procedure reviewed**	**Previous CSST review ID**
SCML snapshot ID	**Analysis file (if used)**	**Prepared by / role**	**Controlled location**

A Purpose and use

The CSST is the site's review of its complications taken together. A single event is assessed in the SART and indexed in the SCML; the CSST looks at all of them at once, on the site's own schedule and after a trigger, to see whether events that each seemed explicable share a cause: the same component, the same surgeon, the same moment of follow-up or the same kind of case. A concentration of events is a question to test, not a conclusion about a device or a surgeon.

Unit of analysis	Decision supported	Inputs	Outputs
One review of the site's complications for one period, or one review opened by a trigger.	Whether the data show no material pattern, a pattern to watch more closely, a signal to send on, or a result limited by the data.	SCML events, ICDL procedures and context, PFVR follow-up reached, SLCT surgeon sequence, SART event status and the previous CSST.	Counts by time point (C) Triggers met (D) Signals assessed (F) Routes to the receiving records (H)

B Sources and their limitations

With limitation means the review goes ahead: name the records that are missing or incomplete. Not available means the counts that need that source cannot be made in this review.

Source	Used for	ID · extraction date	Status	Limitation (records missing or incomplete)
SCML snapshot	Events and their categories		☐ Complete ☐ With limitation ☐ Not available	
ICDL procedure records	Procedures, components, configuration, context		☐ Complete ☐ With limitation ☐ Not available	
PFVR / visit schedule	Participants who reached each follow-up time point		☐ Complete ☐ With limitation ☐ Not available	
SLCT	Surgeon codes and case sequence		☐ Complete ☐ With limitation ☐ Not available	
SART	Event type, seriousness and status		☐ Complete ☐ With limitation ☐ Not available	
Previous CSST review	Signals to follow up in G		☐ Complete ☐ With limitation ☐ Not available	

C1 Cases and follow-up reached

These numbers are the denominators for C2 and C3. Procedures and surgeons come from the ICDL and the SLCT, enrollment from the PSEAR, follow-up from the PFVR. The time points match the columns of C2. Adapt the time points to the CIP visit schedule.

Count	This period (n)	Previous period (n)	Change (%)	Cumulative (N)
Participants enrolled				
Investigational procedures performed				
Surgeons operating				
Surgeons with a first study case in this period				
Procedures within a surgeon's first ___ study cases				
Revision procedures (component exchanged or removed)				
Explanted components returned				
Follow-up visits due				
Follow-up visits done in window				
Follow-up visits missed				
Participants who reached each time point (N for the columns of C2)				
Operated (intraoperative)				
Reached discharge				
Reached first postoperative visit				
Reached 6 weeks				
Reached 3 months				
Reached 6 months				
Reached 12 months				
Reached 24 months				
Participants withdrawn or lost to follow-up				

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Complications by time point · 1 of 2

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C2 Complications by follow-up time point

Count each event once, in the time point where it occurred, against the N of that column. Reading across a row shows whether a problem appears early or grows with time.

Event (SCML category)	Intraoperative (n / N) %	Before discharge (n / N) %	First postop visit (n / N) %	6 weeks (n / N) %	3 months (n / N) %	6 months (n / N) %	12 months (n / N) %	24 months (n / N) %	Total (n)	New this period
N for this column (from C1)										
Instability / dislocation										
Dislocation										
Subluxation or instability symptoms										
Fracture										
Intraoperative fracture										
Periprosthetic fracture										
Acromion or scapular spine fracture										
Infection / wound										
Superficial wound problem										
Deep or periprosthetic infection										
Hematoma										
Neurologic / vascular										
Nerve injury (axillary nerve, brachial plexus)										
Vascular injury										
Device performance										
Component loosening										
Component disassociation (liner, glenosphere)										
Implant breakage or wear										
Radiographic change (notching, lucency, migration)										

C2 Complications by follow-up time point (continued)

Event (SCML category)	Intraoperative (n / N) %	Before discharge (n / N) %	First postop visit (n / N) %	6 weeks (n / N) %	3 months (n / N) %	6 months (n / N) %	12 months (n / N) %	24 months (n / N) %	Total (n)	New this period
N for this column (from C1)										
Function and other										
Stiffness or loss of motion										
Persistent pain										
Thromboembolic event										
Other (SCML code): _____										

Overview

Measure	This period (n)	Previous period (n)	Change (%)	Cumulative (n / N) %	Cumulative at previous review (n / N) %
Participants with at least one complication					
Participants with more than one complication					
Reoperations					
Revisions (component exchanged or removed)					
Serious events (SAE / SADE)					
Device deficiencies (all; counted apart from complications)					

C3 Procedure context

For each marker, compare the complication rate in procedures with it and without it. Context never changes the counts in C2; it helps explain them.

Context marker	Procedures with it (n / N) %	With it: had a complication (n / n) %	Without it: had a complication (n / n) %	Source
Unplanned configuration change				
Component substitution				
Technical deviation from the surgical technique				
Intraoperative difficulty (bone quality, deformity, revision case)				
Senior support or proctoring used				
Surgeon within first ___ study cases				
Operation outside normal hours				
Instrument or loaner set problem				
Gap in the operative record				
Other: _____				

Links between context and complications

Context marker	Procedures (ICDL IDs)	Complication found (C2 event · SCML ID)	Comment

E Comparison by surgeon, component and case type

A group stands out when its share of complications is clearly larger than its share of procedures. Use the surgeon codes of the SLCT and the component, lot and configuration recorded in the ICDL.

E1 · By surgeon (SLCT code)

Surgeon	Procedures (n) % of all	With a complication (n / n) %	Share of all complications %	In the first ___ cases (n / n) %	Note

E2 · By component, lot or configuration (ICDL)

Component · lot · configuration	Procedures (n) % of all	With a complication (n / n) %	Share of all complications %	Note

E3 · By case type

Case type (indication, primary or revision)	Procedures (n) % of all	With a complication (n / n) %	Share of all complications %	Note

D Signal triggers

Apply each trigger and write the value found. A trigger met opens a signal in F.

ID	Trigger	What is compared · where	Agreed limit	Source of the limit	Value found (n / N) %	Met?
T1	**Rate above the agreed limit**	Any category at any time point (C2).				☐ Yes ☐ No ☐ Not assessable
T2	**Rate rising with time**	The same category grows from earlier to later time points (C2 row).				☐ Yes ☐ No ☐ Not assessable
T3	**Rise since the previous review**	Same category and time point, compared with the previous CSST.				☐ Yes ☐ No ☐ Not assessable
T4	**New category**	A category with no earlier events now has events (C2, New this period).				☐ Yes ☐ No ☐ Not assessable
T5	**Surgeon share**	A surgeon's share of complications exceeds their share of procedures (E1).				☐ Yes ☐ No ☐ Not assessable
T6	**Component or configuration share**	Same comparison by component, lot or configuration (E2).				☐ Yes ☐ No ☐ Not assessable
T7	**Early learning cases**	Complication rate in a surgeon's first cases above their later cases (E1, C1).				☐ Yes ☐ No ☐ Not assessable
T8	**Context association**	Rate with a context marker above the rate without it (C3).				☐ Yes ☐ No ☐ Not assessable
T9	**Serious events grouped**	Serious events in one category within a short window (SART, C2).				☐ Yes ☐ No ☐ Not assessable
T10	**Device deficiencies of one type**	Repeated deficiencies of the same type or component (C2, SART).				☐ Yes ☐ No ☐ Not assessable

These triggers are examples. Before the first inclusion, review them and set each limit with the site's clinical team and, where it applies, the sponsor; write the agreed limit and its source (CIP, sponsor guidance, site procedure). A limit changed later applies only to later reviews.

F Assessment of each signal

One row per trigger met. Describe the pattern and the explanations still to test; do not conclude the cause. Set how the next review will show whether the response worked.

Signal ID	Trigger (D) · events (SCML IDs)	Pattern described	Explanations to test	Disposition	What will show the response worked
			☐ Device, component or lot ☐ Procedure or technique ☐ Surgeon or learning curve ☐ Case complexity or participant ☐ Data or documentation ☐ Local process ☐ Undetermined	☐ No material pattern ☐ Closer surveillance ☐ Sent on (see H) ☐ Inconclusive: data limitation	
			☐ Device, component or lot ☐ Procedure or technique ☐ Surgeon or learning curve ☐ Case complexity or participant ☐ Data or documentation ☐ Local process ☐ Undetermined	☐ No material pattern ☐ Closer surveillance ☐ Sent on (see H) ☐ Inconclusive: data limitation	
			☐ Device, component or lot ☐ Procedure or technique ☐ Surgeon or learning curve ☐ Case complexity or participant ☐ Data or documentation ☐ Local process ☐ Undetermined	☐ No material pattern ☐ Closer surveillance ☐ Sent on (see H) ☐ Inconclusive: data limitation	
			☐ Device, component or lot ☐ Procedure or technique ☐ Surgeon or learning curve ☐ Case complexity or participant ☐ Data or documentation ☐ Local process ☐ Undetermined	☐ No material pattern ☐ Closer surveillance ☐ Sent on (see H) ☐ Inconclusive: data limitation	

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Earlier signals and routes

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G Signals from earlier reviews

Check each earlier signal against the criterion set for it in F.

Signal ID · review	Criterion set then	Result in this review	Outcome	Next step
			☐ Effective ☐ Partly effective ☐ Not effective ☐ Not assessable	
			☐ Effective ☐ Partly effective ☐ Not effective ☐ Not assessable	
			☐ Effective ☐ Partly effective ☐ Not effective ☐ Not assessable	
			☐ Effective ☐ Partly effective ☐ Not effective ☐ Not assessable	

H Where each signal goes

The action is followed in the receiving record; this review records only that it was sent.

Route	Question for this review	Signal ID (F)	Receiving record · date sent	Status
SLCT	Could surgeon-dependent variation explain the pattern?			☐ Sent ☐ Not required ☐ Pending
Sponsor device-deficiency route	Does the signal point to a device, component, configuration or lot?			☐ Sent ☐ Not required ☐ Pending
SART	Are the individual events complete and routed through the safety pathway?			☐ Sent ☐ Not required ☐ Pending
SRR	Does the signal show a local process weakness to enter as a risk?			☐ Sent ☐ Not required ☐ Pending
CAPA log	Did a first response fail, or does a process failure recur?			☐ Sent ☐ Not required ☐ Pending
POT	Does the PI need to restrict, supervise more, continue or escalate?			☐ Sent ☐ Not required ☐ Pending
PIR / IMC	Should the result inform an indicator or an internal check?			☐ Sent ☐ Not required ☐ Pending

| Conclusion and sign-off

The conclusion reflects the most serious open signal.

Review conclusion	☐ No material pattern ☐ Closer surveillance ☐ Signal sent on ☐ Inconclusive: data limitation
Next review · date and what will be checked again	

Limitations and rationale

Reviewer	**PI / accountable approver**
Name / role:	Name / role:
Signature / date:	Signature / date: